Signals and sampling

- A continuous signal x(t) is sampled every T seconds.

- The sampling rate is f_s = 1 / T.

- Nyquist: f_s must exceed twice the highest frequency.

- Below it, frequencies fold back: aliasing.

Quantization

- Each sample is rounded to one of 2^N levels.

- The step is the full range divided by 2^N.

- Quantization noise is about 6 dB lower per bit.

- Dither spreads the error into broadband noise.

The z-transform

- X(z) is the sum of x[n] z^-n over every n.

- A delay of one sample multiplies by z^-1.

- Poles inside the unit circle mean a stable filter.

- Zeros on the circle remove a frequency.

Filters in practice

- FIR filters are always stable and can have linear phase.

- IIR filters reach a steep edge with fewer taps.

- Windowing trades the main lobe for side lobes.

- Exercise: design a 5-tap moving average.